Active Transport

Active transport is mediated by a protein called as SGLT1 (Sodium Glucose Transporter 1) in the luminal side and GLUT 2 on the basolateral side (Figure 2.14). The events involved in the transport are:

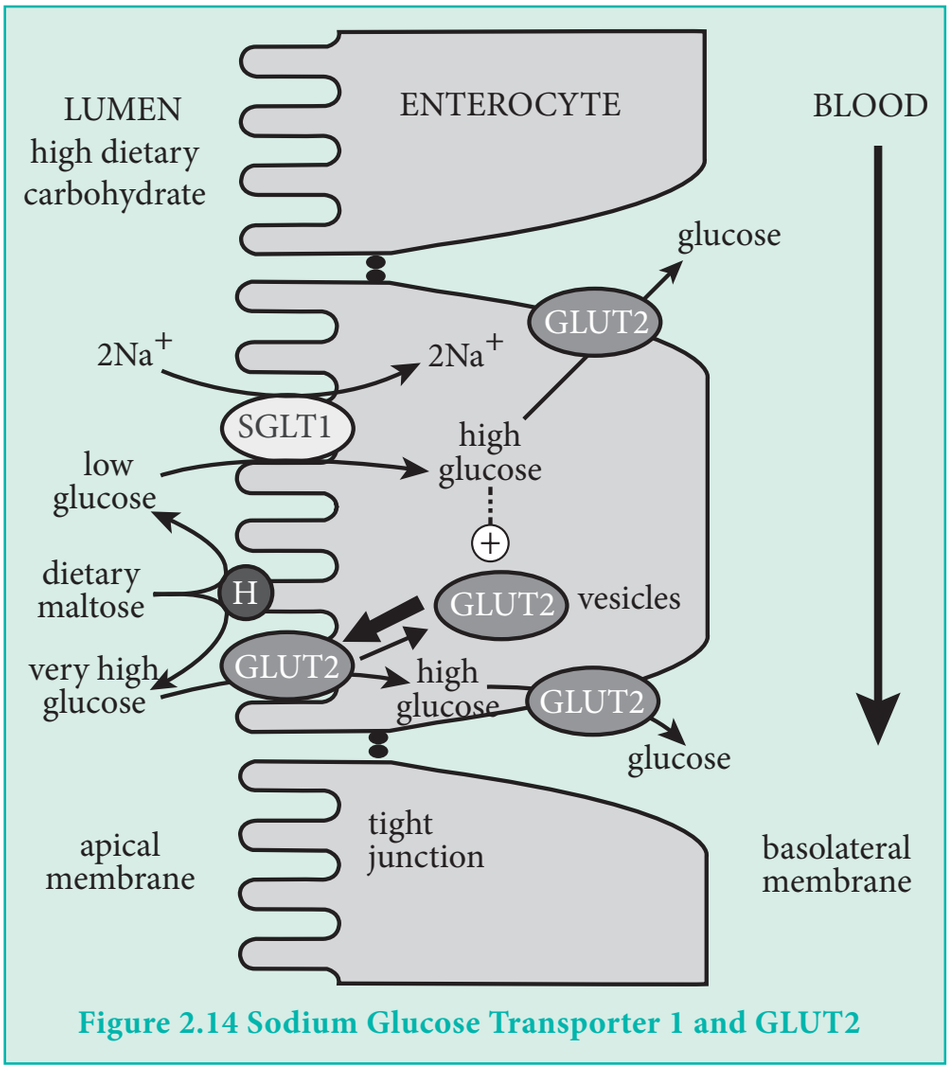


Figure 2.14 Sodium Glucose Transporter 1 and GLUT2

1. A high concentration of K^+ and low concentration of Na^+ is maintained inside the intestinal brush border membrane enterocytes with the help of Na^+-K^+ ATPases, which utilize ATP for its pumping action.
2. SGLT1 is a symporter, which can bind Na^+ and glucose from the gut lumen. With the energy obtained by downhill transport of sodium, glucose is transported in a uphill manner, ie.against the concentration gradient, resulting in high concentrations of glucose inside the enterocyte. The steps involved in this transport are :

- The transporter is initially facing the lumen and oriented such that it can bind sodium and not glucose.
- After the binding of sodium, a conformational change is induced that opens the glucose-binding pocket
- Glucose binding induces a conformational change so that the transporter now faces the cytosol.
- Sodium and glucose dissociate into the cytoplasm
- After delivering the cargo, the transporter reorients back to the luminal side.

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